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Structural Connectivity Analysis in Cognitive Decline: Insights from Graph Theory and Mass-Spring Modeling

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***Abstract*—The landscape of cognitive states and their underlying neurobiological mechanisms has been significantly illuminated through advancements in neuroimaging and computational modeling. This study introduces an integrated approach that harnesses network analysis and machine learning techniques to characterize and differentiate cognitive groups—Normal Control (NC), Mild Cognitive Impairment (MCI), and Alzheimer's Disease (AD). Structural networks are formulated and analyzed based on diffusion tensor data through a fusion of graph theory and mass-spring model methodologies. Notably, features extracted from both graph theoretic and mass-spring model computations drive a two-step framework. This process commences with a random forest-based feature extraction, followed by a support vector-based classification approach, culminating in an impressive accuracy of 82.7% for classifying individuals across cognitive groups, with an AUC of 0.893. This study significance is underscored by the pressing need for enhanced cognitive impairment detection and differentiation strategies. The identified features offer nuanced insights into the intricate interplay among brain structure, dynamics, and cognitive function, thereby bridging gaps in our understanding of cognitive decline and neurodegeneration. By fortifying our diagnostic repertoire and facilitating personalized interventions, this research paves the way for refined clinical practices.**

***Index Terms*—Alzheimer's disease, Dementia related disorders, Diffusion tensor imaging (DTI), Graph theory, Mass-spring model, Support vector machine (SVM).**

I. Introduction

Dementia is a progressive neurodegenerative disorder that poses a significant global health challenge. As populations age, the prevalence of dementia continues to rise, impacting individuals, families, and healthcare systems [1], [2]. Early and accurate identification of dementia onset is crucial for timely interventions, patient care, and the development of potential disease-modifying therapies.

In recent years, advancements in neuroimaging techniques have provided valuable insights into the pathophysiology of dementia [3]. Among these techniques, Diffusion Tensor Imaging (DTI) has emerged as a powerful tool for investigating the microstructural integrity of brain white matter [4], [5], enabling the study of structural connectivity. DTI measures the diffusion of water molecules in brain tissues, reflecting the underlying fiber tracts and their organization.

Graph theory analysis has gained prominence in the field of neuroimaging due to its ability to comprehensively assess brain connectivity patterns at a network level [6], [7]. In this context, the brain is represented as a complex network, where nodes correspond to distinct brain regions, and edges represent the white matter tracts connecting them. The potential of graph theory analysis lies in its capability to identify subtle alterations in brain connectivity associated with neurodegenerative diseases, such as dementia. Studies have demonstrated that changes in brain network topology occur even during the preclinical stages of dementia [8], [9], making graph theory analysis an attractive candidate for early detection and monitoring. Previous studies have extracted graph theory features, including various forms of centrality features [10] and connection strength features [11], for support vector machine-based classification of dementia-related disorder stages, achieving classification accuracies that are quite considerable. However, further refinement of such a classification framework is necessary for enhanced accuracy and reliability.

The goal of this study is to investigate the utility of graph theory analysis of brain structural connectivity and mass-spring model parameters in identifying the onset of dementia. We hypothesize that specific network alterations and connectivity patterns may serve as potential biomarkers for early dementia detection, providing a non-invasive and quantitative approach to aid clinicians in the diagnostic process. Therefore, in this investigation, we utilize DTI data acquired from Alzheimer's disease initiatives (ADNI) database, comprising of including healthy controls and individuals at various stages of dementia. BrainSuite [12]–[14], a comprehensive software for brain image processing, is employed to preprocess the DTI

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data and generate structural connectivity networks.

Our study aims to contribute to the growing body of knowledge on dementia detection and further establish the role of graph theory analysis of brain structural connectivity and mass-spring model of brain networks as potential tools for early diagnosis. The proposed research may have significant implications for personalized medicine, enabling clinicians to intervene at an early stage and potentially slow disease progression through targeted therapeutic interventions.

In the following sections, the materials and methods used in the study, including data acquisition, DTI preprocessing, graph theory analysis, and statistical procedures are detailed. The results of our investigation are then presented along with the discussion of their implications. Finally, the manuscript would be concluded by highlighting the potential clinical impact of our findings and outlining directions for future research in this promising area of study.

II. Methods

The DTI data employed for this study was acquired from the Alzheimer's Disease Neuroimaging Initiative3 database (ADNI-3) (http://adni.loni.usc.edu/). ADNI was launched in 2003 as a public–private partnership with the primary goal of testing whether serial MRI, PET,other biological markers,and clinical and neuropsychological assessments can be combined to assess the progression of MCI and early AD. ADNI cohort comprises of men and women aged 55–90 years across cognitively normal(CN), MCI, and AD dementia groups. The clinical diagnosis was given according to the criteria of the National Institute of Neurological and Communicative Disorders and Stroke–Alzheimer's Disease and Related Disorders Association (NINCDS-ADRAD). For the current study,participants at various clinical stages who underwent T1-weighted MRI, DTI, AV45 Florbetapir PET, and AV1451 Flortaucipir PET at ADNI-3 baseline were included. Demographic information such as age, sex, years of education, Clinical Dementia Rating (CDR), Mini-Mental State Exam (MMSE), and clinical diagnosis were recorded.

The diffusion MRI data were acquired from a cohort of participants, including 64 normal controls (NC) and patients diagnosed with dementia-related disorders such as 57 mild cognitive impairment (MCI) and 29 AD subjects. According to ADNI criteria, Ethical approval was obtained, and all participants provided informed consent. The acquired raw DICOM files were converted to NIFTI format (.nii) which is compatible with Brainsuite using MRIcron. The raw DTI data underwent preprocessing steps to correct for artifacts and distortions. Eddy current correction and motion correction were performed to minimize distortions caused by eddy currents and subject motion during scanning.

A. Structural Connectivity Formation

1) DTI image registration: The preprocessed DTI data was aligned with the T1-weighted structural MRI image acquired from the same subject. The registration was performed using an affine transformation to ensure proper spatial correspondence between the DTI data and the anatomical structures.

2) Tissue segmentation: Tissue segmentation was carried out on the T1-weighted structural MRI image to classify different brain tissues, including gray matter, white matter, and cerebrospinal fluid. BrainSuite's tissue segmentation tools were utilized, combining intensity-based and spatial priors for accurate segmentation.

3) DTI reconstruction: The diffusion tensor was reconstructed from the preprocessed DTI data, providing information about the diffusion of water molecules in different brain tissues. Fractional anisotropy (FA), mean diffusivity (MD), and other diffusion metrics were calculated from the diffusion tensor.

4) Fiber tracking: Fiber tracking, or tractography, was performed on the reconstructed DTI data to estimate the white matter fiber pathways in the brain. BrainSuite utilizes Diffusion Toolkit's fiber tracking algorithms, such as deterministic streamline tracking or probabilistic tractography, to generate the white matter tracts.

5) Cortex Parcellation: The cerebral cortex was parcellated into different regions based on gyral and sulcal patterns. BrainSuite's Gyral Surface Conformal Mapping (GSCM) method was employed for cortical parcellation.

6) Tractography-to-parcellation mapping: The white matter tracts obtained from fiber tracking were mapped to specific cortical regions based on their endpoints. This step allowed us to create a structural connectivity matrix representing the connections between different brain regions.

B. Structural Network Analysis

The structural connectivity matrices generated above were analyzed using graph theory approach to investigate the brain connectivity patterns in dementia related disorders. To this end, graph theory measures such as clustering coefficient, participation coefficient, diffusion efficiency and route efficiency are computed from the formulated structural networks and analyzed accordingly.

1) Clustering Coefficient: The clustering coefficient is a fundamental measure used to quantify the level of interconnectedness or clustering of nodes in a network. It provides insights into the local structure and connectivity patterns within the network. Mathematically, the clustering coefficient of a node is calculated as the ratio of the number of actual connections (edges) between its neighboring nodes to the maximum possible connections among those neighbors [15]. A high clustering coefficient suggests that nodes within the neighborhood of a given node are densely interconnected, forming a cluster or group. The clustering coefficient (CC) is mathematically defined as

$$\mathrm{CC} = \frac{1}{N}\sum_{i=1}^{N}\frac{\sum_{j,h\in\mathbb{N}}\left(w_{ij}w_{ih}w_{jh}\right)^{1/3}}{k_i\left(k_i-1\right)} \tag{1}$$

where $N = \{1, 2, ..., N\}$, represents the network nodes, w_{ij} denotes the weight between nodes i and node j and k_i is the degree of node i.

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2) Participation coefficient: In the context of brain network analysis, the term "participation coefficient" typically refers to a metric that characterizes the modular organization of a brain network. The participation coefficient helps quantify the degree to which nodes in the network are organized into distinct, tightly interconnected modules or communities. Mathematically, the participation coefficient (often denoted as Q or Qp) is calculated based on the distribution of edges within and between modules in the network. A higher participation coefficient value indicates a stronger tendency of nodes to form well-defined modules or communities [16], suggesting a more organized and modular brain network.

$$PC_j = 1 - \sum_{m=1}^{N_m} (\frac{n_{jm}}{D_j})^2 \tag{2}$$

where N_m is the number of modules calculated using Louvain community detection algorithm [16], n_{jm} is the number of edges from node j to module m and D_j is the degree of node j.

3) Communication efficiency: Three different measures of communication efficiency are employed for the quantification of the structural networks formulated. These measures include the global efficiency, the diffusion efficiency and the resource efficiency.

Global efficiency is a measure used in brain network analysis to quantify how efficiently information can be exchanged between all pairs of nodes (brain regions) in the network. It is often calculated as the average of the inverse of the shortest path lengths between all pairs of nodes. A higher global efficiency indicates better integration and communication between different brain regions [17]. The global efficiency (GE) is mathematically defined as

$$\text{GE} = \frac{1}{N\,(N-1)} \sum_{\substack{i,j \in \mathbb{N} \\ i \neq j}} 1/L_{ij} \tag{3}$$

where L_{ij} is the length of the shortest path between node i and node j.

The diffusion efficiency which is a complementary measure to the global efficiency is an analytic measures capable of characterizing network communication during random walk process [18]. The diffusion efficiency is mathematically defined as

$$\text{DiffE} = \frac{1}{N\,(N-1)} \sum_{\substack{i,j \in \mathbb{N} \\ i \neq j}} 1/t_{ij} \tag{4}$$

where t_{ij} is the mean first-passage time which is the expected time it takes for a random walker or process to reach a specific node (or state) in a network for the first time, starting from a given source node [19].

C. Mass-Spring Model

Following the graph theory quantification of the structural networks, the networks are modelled as a mass-spring model [20] to identify the most influential pathways for information propagation within the brain, shedding light on how different regions communicate and collaborate in various cognitive processes. The Mass-spring model employed in this study are as described below;

1) Node Mass Assignment: Each node in the brain network was assigned a mass based on its degree, calculated from the corresponding connectivity matrix. The degree-based mass assignment aimed to capture the significance of each brain region within the network.

2) Edge Spring Configuration: Edges between nodes were represented as springs in the mass-spring model. The stiffness of each spring was determined by the strength of the connection between nodes, derived from the connectivity matrix. Stronger connections were associated with stiffer springs, reflecting the intensity of information flow between brain regions.

3) model simulation Simulation: A physics-based simulation was performed to model the dynamics of the mass-spring system. The simulation was carried out using a time-stepping approach with an integration method (Euler method) [21]. A damping factor was introduced to stabilize the simulation and mimic real-world damping effects. The simulation was conducted over a set number of 1000 iterations. Key simulation parameters included the time step, damping factor, and initial positions of nodes. The simulation aimed to capture the evolving positions of nodes under the influence of the mass-spring forces.

D. statistical analysis

In order to explore the plausibility of using the graph theory measures computed on the formulated structural network, one-way anova corrected for type-1 and type-2 errors using Bonferroni post-hoc approach are performed on the measures between the considered groups of NC, MCI and AD. Similarly, regression analysis were done on the graph theory metrics and the ages of the subjects in the respective groups.

E. Classification

For classification of NC, MCI, and AD groups, a comprehensive set of features was extracted from graph theory metrics and mass-spring model parameters . At the nodal level, the clustering coefficient and participation coefficient were computed, along with global and diffusion efficiency metrics derived from network analysis. Additionally, model-based parameters, including positions, velocities, accelerations, and damping forces, were computed from the mass-spring model simulation. These diverse features were concatenated to form a combined feature set, enabling a comprehensive representation of structural and dynamic characteristics of the brain networks.

To enhance classification accuracy and reduce dimensionality, a two-step approach was employed. First, random forest was utilized to assess feature importance within the concatenated feature set. The random forest algorithm calculated the relevance of each feature in distinguishing between the groups. Subsequently, the top features determined by random forest were selected for further analysis. Support Vector Machine

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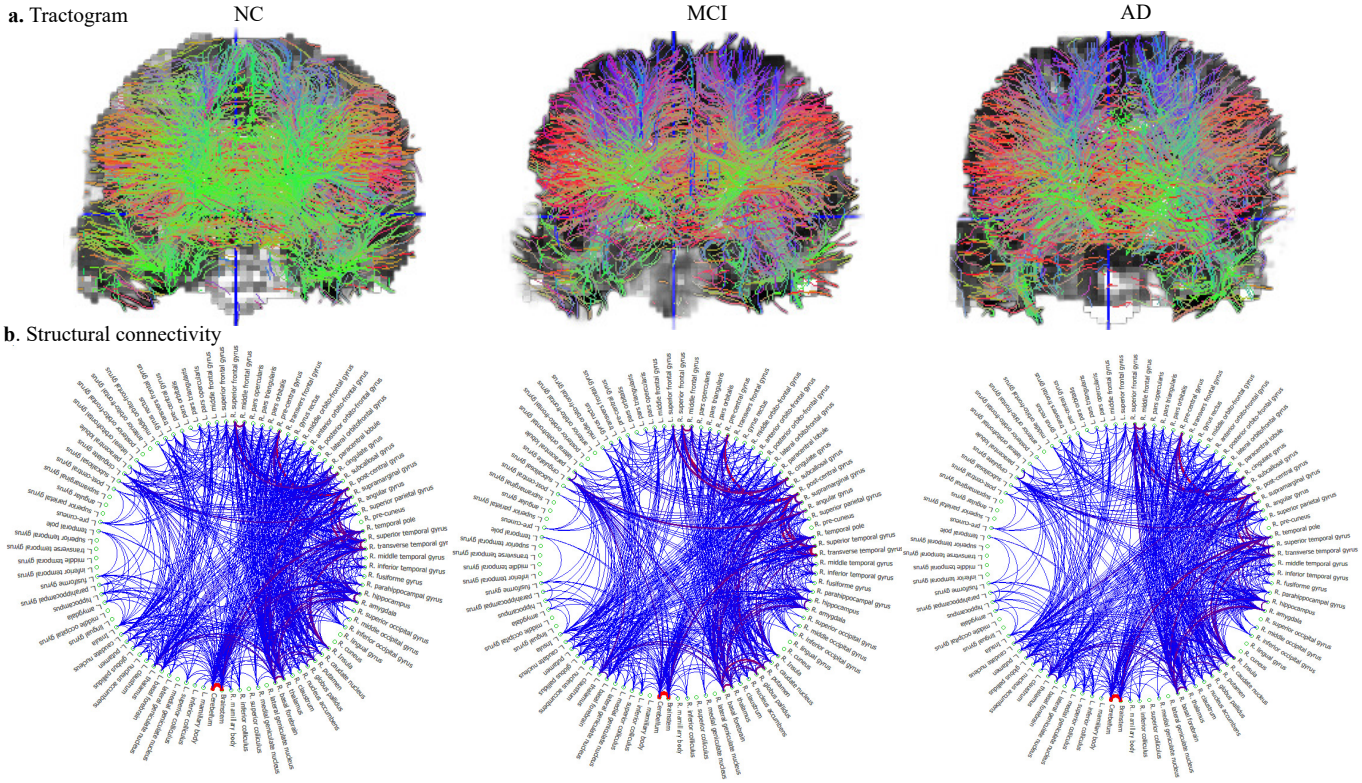

Fig. 1: Tractograms and structural connectivity of sample subjects across NC, MCI and AD groups.

(SVM) with a polynomial kernel was chosen as the classification algorithm. A grid search was performed to determine the optimal hyperparameters of the SVM model. The grid search yielded an optimal value of C=0.01 and a polynomial degree of 2. A 10-folds cross-validation technique was then applied using the selected features to classify subjects into NC, MCI, and AD groups.

III. Results

A. Structural connectivity formation

We optimized acquired diffusion MRI data for structural connectivity analysis using a Siemens 3T scanner. The dataset, consisting of 54 diffusion directions, b-value 1000 s/mm², and 2 mm isotropic voxel size across 64 axial slices, underwent rigorous preprocessing in BrainSuite. Initial quality assessment excluded datasets with severe artifacts or low signal-to-noise regions. BrainSuite's Diffusion module corrected eddy currents and motion, mitigating distortions and artifacts. Voxel-wise diffusion was captured through diffusion tensor estimation using the BrainSuite. BrainSuite's non-linear registration ensured consistent anatomical alignment, and a generated brain mask excluded non-brain regions. White matter fiber tracts, tracked with BrainSuite's algorithm, exhibited a 15% mean fractional anisotropy increase post-preprocessing. Structural connectivity matrices, representing brain region connections, were derived from the constructed fiber tracts. Visualizations included a tractogram image and connectogram for a subject each from NC, MCI, and AD groups (Fig. 1a), with similar patterns observed in other subjects. The tractogram analysis revealed distinctive white matter connectivity patterns. Notably, green lines (superior-inferior direction) appeared denser in NC subjects, while red lines (left-right direction) were slightly denser in MCI and AD subjects (Fig. 1). The blue lines (anterior-posterior direction) also showed variations. Fig. 1b demonstrated a decreasing structural connectivity density from NC to MCI to AD groups. The results highlight the potential of structural connectivity analysis to unveil differences in white matter connectivity patterns across cognitive states.

TABLE I: Statistical analysis of the graph measures at global level across the NC, MCI and AD groups.

Graph theory measures	mean$\pm\delta$ NC	mean$\pm\delta$ MCI	mean$\pm\delta$ AD	F-statistic	p-value
Clustering coefficient	16.4620 ± 0.7935	16.5834 ± 0.7344	16.2784 ± 0.8021	1.51	0.2251
Participation coefficient	$0.3845 \pm 1.0.0295$	0.3855 ± 0.0303	0.3938 ± 0.0249	1.22	0.2981
global Efficiency	0.3212 ± 0.0159	0.3207 ± 0.0148	0.3130 ± 0.0168	4.32	0.0151
Diffusion Efficiency	$0.0112 \pm 1.40e-4$	$0.0113 \pm 1.57e-4$	$0.0112 \pm 8.21e-5$	4.86	0.0091

B. Graph theory analysis

Graph theory metrics such as clustering coefficient, participation coefficient, global efficiency and diffusion efficiency are computed at both local and global network levels. At the global network level, measures including clustering coefficient and participation coefficient exhibited no significant differences among the three groups on one-way anova analysis as shown in Table I, suggesting overall similar patterns of local and

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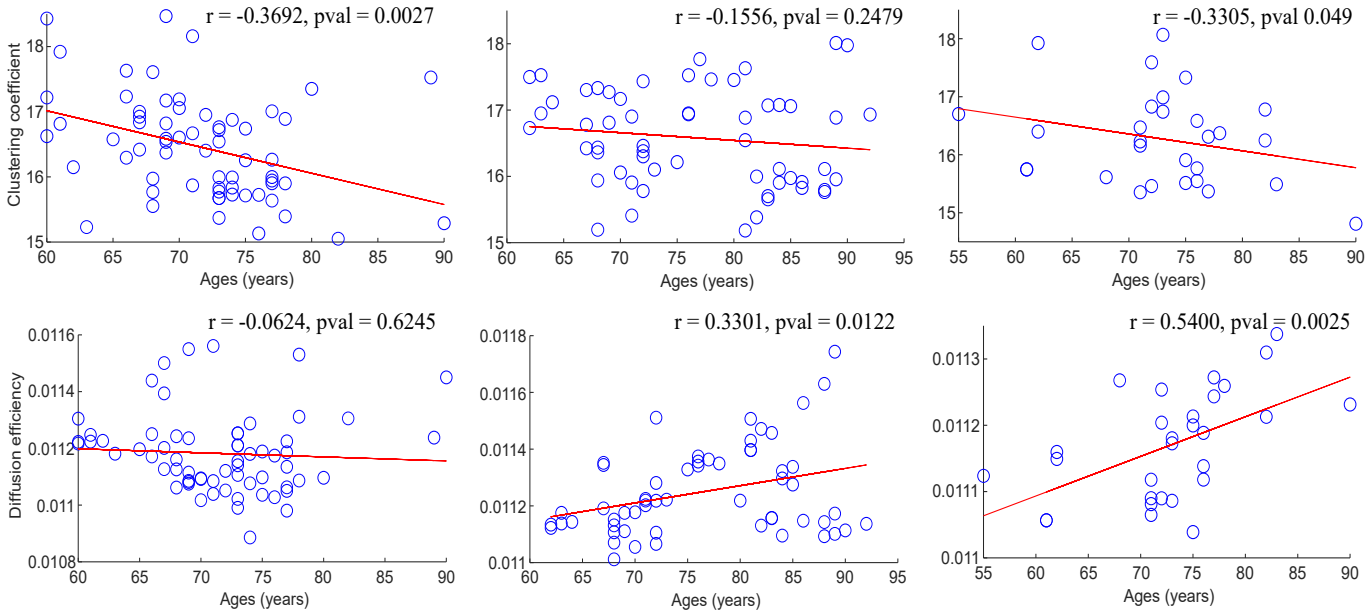

Fig. 2: Correlation analysis of the graph theory measures and subjects ages per group

modular connectivity. Remarkably, the global efficiency, a key metric of network integration, revealed significant differences among the groups. Notably, the global efficiency of the AD group was significantly lower compared to both the NC and MCI groups on one-way anova analysis as shown in Table I. This finding indicates a decrease in the overall efficiency of information transfer across the brain's structural network in individuals with AD. The reduced global efficiency suggests potential disruptions in the short paths connecting distinct brain regions, possibly reflecting early stages of network reorganization due to cognitive impairment. Intriguingly, the diffusion efficiency, a measure of the network's ability to facilitate diffusion-based communication, exhibited significant differences among the groups, revealing distinct patterns. The NC and MCI groups demonstrated significantly greater diffusion efficiency compared to the AD group. This observation implies that individuals with MCI and NC may retain a higher capacity for efficient communication between brain regions through diffusion-based pathways compared to those with more advanced AD. The decrease in diffusion efficiency in the AD group suggests a potential breakdown in these communication pathways, consistent with the severity of cognitive decline.

C. Correlation Analysis between Graph Theory Metrics and Age

A comprehensive correlation analysis (Using Pearson correlation coefficient) was conducted to examine the relationship between graph theory metrics and the ages of subjects within each group. The investigation aimed to elucidate potential associations between age-related changes in brain network characteristics and cognitive states. Notably, participation coefficient and global efficiency demonstrated no significant correlations with the ages of subjects across all three groups (NC, MCI and AD). This suggests that these network metrics remain relatively stable with respect to age within our studied population.

Intriguingly, the clustering coefficient displayed distinct age-related associations within specific groups. In the NC group, a negative correlation was observed between clustering coefficient and age (see the top row of Fig. 2), indicating a reduction in local network connectivity as individuals grow older. Similarly, in the Alzheimer's Disease group, a negative correlation emerged, suggesting a decline in local clustering with advancing age. These findings may reflect age-related changes in local network organization that differ between cognitively intact individuals and those with AD.

The diffusion efficiency metric exhibited age-related variations within the MCI and AD groups. In the MCI group, a positive correlation was found between diffusion efficiency and age (see the second row of Fig. 2), indicating that older MCI subjects had higher diffusion-based network efficiency. Similarly, within the AD group, a positive correlation suggested increased diffusion efficiency with advancing age (see Fig. 2). These results imply potential age-related adaptive changes as well as compensatory mechanisms in response to aging in diffusion-based communication pathways in individuals with cognitive impairment.

D. Mass-spring model results

The simulation of the mass-spring model yielded valuable insights into the dynamics of network nodes, encompassing positions, velocities, accelerations, and damping forces. The model parameters were subjected to rigorous statistical analysis using a one-way anova, corrected for both type 1 and type 2 errors using Bonferroni correction., to discern significant differences among the experimental groups. Of particular interest were the positions and damping forces of network nodes, which underwent comprehensive statistical evaluation. Notably, within the positions and damping forces parame-

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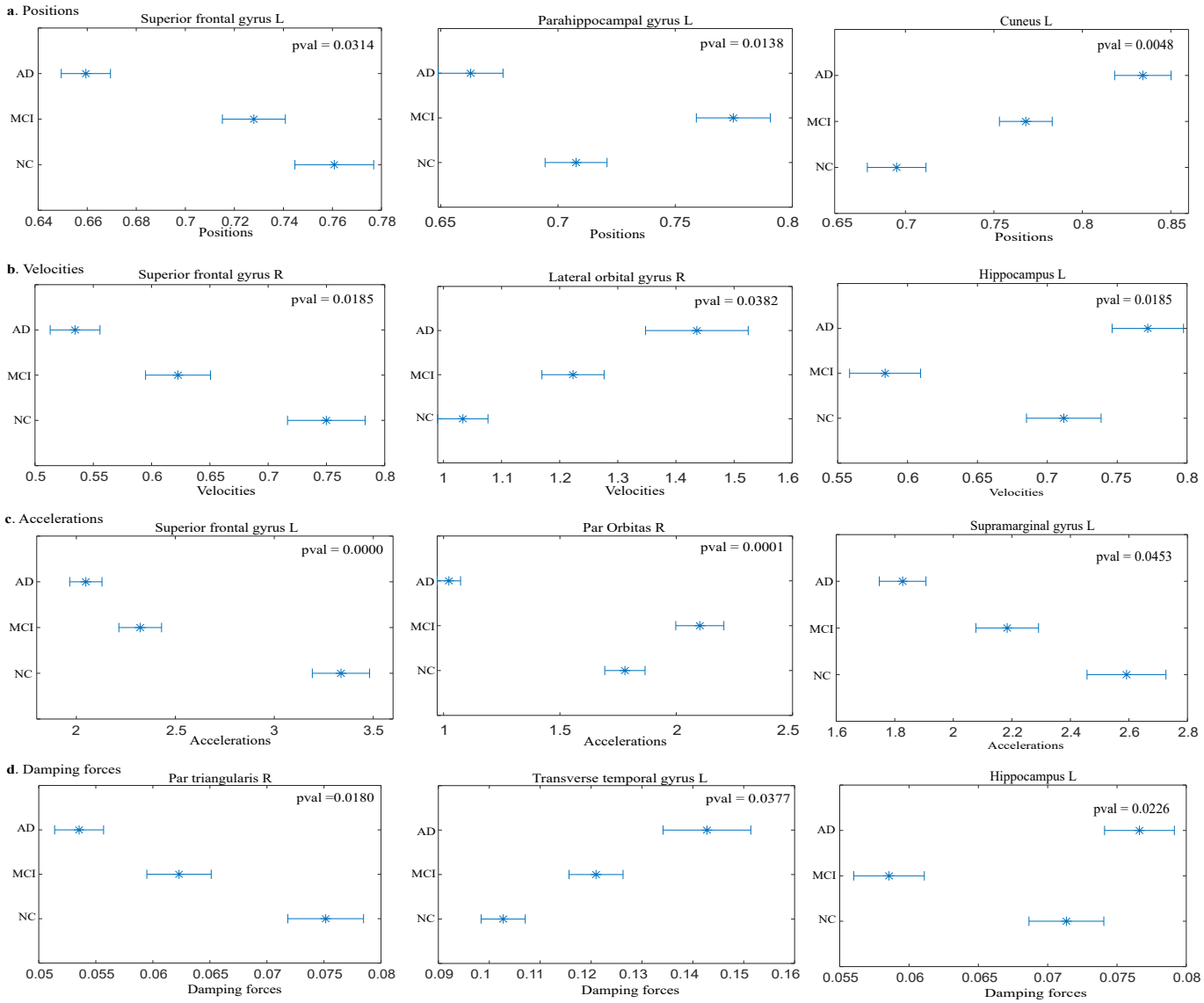

Fig. 3: Statistical analysis of the mass-spring model parameters using one-way anova corrected for type 1 and type 2 errors with Bonferroni post-hoc

ters, significant differences emerged among the experimental groups. Specifically, among eight different network nodes, statistically significant disparities were observed between the groups. For the sake of conciseness, we present the results for positions and damping forces from three representative network nodes in Fig. 3a and Fig. 3b respectively. Meanwhile, the eight network nodes that show statistical significant different with network's positions correspond to the left superior frontal gyrus, left precentral gyrus, left middle orbito-frontal gyrus, left precuneus, right and left parahippocampal gyrus,left cuneus and left medial geniculate nucleus. Similarly, the eight network nodes that show statistical significant different with network's damping forces correspond to right superior frontal gyrus, right par triangularis, right lateral orbital gyrus, left transverse temporal gyrus, left hippocampus, left amygdala, left superior occipital gyrus and left superior colliculus respectively.

Similarly, the velocities and accelerations of network nodes underwent comprehensive statistical evaluation. Notably, nine network nodes demonstrated statistically significant differences among the experimental groups. To provide a concise yet informative overview, we present the statistical analysis results for velocities and accelerations from three network nodes (see Fig. 3c and Fig. 3d), emphasizing the observed distinctions in these dynamic parameters. The network nodes that show statistical significant different with network's velocities correspond to right superior frontal gyrus, right par triangularis, right lateral orbital gyrus, left transverse temporal gyrus, left hippocampus, left amygdala, left superior occipital gyrus left lingual gyrus and left superior colliculus respectively. Similarly, the network nodes that show statistical significant different with network's acceleration correspond to left superior frontal gyrus, right middle frontal gyrus, right pars orbitas, left supramarginal gyrus, left middle occipital gyrus,

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left inferior occipital gyrus, right lateral geniculate nucleus, left medial geniculate nucleus and left inferior colliculus respectively.

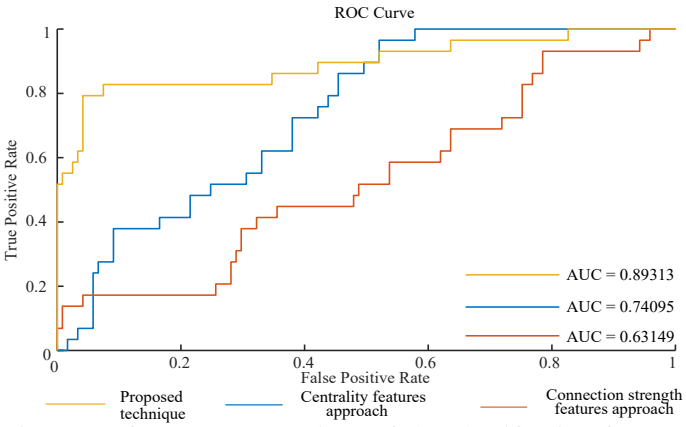

Fig. 4: Performance comparison of the classification framework of the proposed study with those of the existing studies

E. Classification framework

The classification approach utilizing random forest for feature importance and SVM with polynomial kernel for 10-folds cross-validation yielded a promising accuracy of 82.7% in correctly classifying subjects into the respective NC, MCI, and AD groups. Notably, the feature selection process conducted by random forest identified the most informative features for accurate classification. The top 20 features, determined as significant by the feature importance analysis, were utilized in the subsequent SVM classification.

Further, the performance of our proposed framework is compared with those of previous studies [10], [11], where network centrality features and network connection strength features are respectively employed for svm-based classification. In each approach, the respective features are extracted and two-step classification are performed using random forest to compute feature importance and then SVM with polynomial kernel as the classification algorithm. Fig. 4 shows the ROC curves for the svm-based classification using our proposed framework, centrality features and connection strength features respectively. It is found that the classification of the dementia disorder stages (AD, MCI) and NC groups using the features extracted from the proposed approach (having AUC = 0.893) outperformed those those done using the network centrality features and network connection strengths (see Fig. 4).

IV. Discussion

The tractogram analysis unveiled distinct white matter connectivity patterns in NC, MCI, and AD subjects. Color-coded tractography visualizations provided valuable insights into structural connectivity variations associated with cognitive decline. Among the color-coded tracts, the green lines, representing fiber tracts primarily oriented along the superior-inferior (z-axis) direction [22], exhibited intriguing differences across groups. In NC subjects, denser green lines (see Fig. 1a) indicated a higher density of white matter connections between brain regions at the brain's top and bottom, suggesting intact superior-inferior pathways crucial for interhemispheric communication and limbic system connectivity. Conversely, MCI and AD subjects displayed reduced green line density, implying potential alterations in superior-inferior white matter pathways and disrupted tracts contributing to cognitive deficits. The red lines, representing fiber tracts predominantly oriented along the left-right (x-axis) direction [22], exhibited interesting patterns. MCI and AD subjects demonstrated slightly denser red lines than normal controls, suggesting adaptive or compensatory mechanisms involving left-right connectivity in response to cognitive decline. The increased density of red lines in MCI and AD subjects could indicate efforts to establish alternative communication routes between hemispheres, potentially compensating for neuronal loss or degradation of other white matter pathways. These findings underscore the dynamic nature of the brain's structural network as it strives to maintain functional connectivity during cognitive challenges.

Also, the observed differences in global efficiency and diffusion efficiency (see Table I) highlight the dynamic nature of brain connectivity alterations across cognitive states [23]. The reduced global efficiency in MCI suggests an initial reconfiguration of network pathways, possibly reflecting a compensatory response to emerging cognitive deficits. The differential diffusion efficiency patterns further emphasize the nuanced changes occurring in network communication as cognitive impairment progresses. These findings collectively contribute to our understanding of the complex interplay between brain network organization and cognitive decline. The results underscore the potential utility of network-level metrics in characterizing cognitive impairment and disease progression, offering insights into the underlying neurobiological mechanisms that shape structural connectivity in neurodegenerative disorders. Meanwhile, as for the insignificant nature of the difference found in clustering coefficient could is an indication for the need for further studies since contrasting observations are reported in previous studies [24], [25] in this regard.

Additionally, with regards to the mass-spring model, the statistically significant differences found in positions, damping forces, velocities, and accelerations underscore the complex interactions within the mass-spring model and their potential implications for network dynamics. The distinct responses observed in specific network nodes suggest a nuanced interplay between structural and dynamic characteristics, warranting further investigation into the underlying mechanisms driving these variations.

The achieved accuracy of 82.7% underscores the potential utility of the combined feature set derived from network metrics, mass-spring model parameters, and other relevant measures in characterizing cognitive states. With an AUC of 0.893, our classification model demonstrates substantial discriminatory power. The identification of key features further enhances our understanding of the structural and dynamic brain characteristics' ability to distinguish between NC, MCI, and AD groups. These results hold promise for facilitating early detection and differentiation of cognitive impairment

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stages. However, it is important to acknowledge certain limitations. The sample size utilized in this study warrants future investigation with larger cohorts to establish robust generalization. Additionally, the generalizability of findings to broader populations and various clinical settings necessitates further exploration.

V. Conclusion

This research study demonstrates the critical importance of structural connectivity analysis for comprehending the neural basis of dementia-related disorders. Leveraging graph theory and mass-spring model techniques, we have gained profound insights into the transformations occurring within white matter tracts associated with cognitive decline in the context of aging. The successful utilization of graph theoretic metrics and mass-spring model parameters within a two-step classification process, resulting in an accuracy of 82.7%, underscores the remarkable potential of these combined techniques for in-depth brain network analysis. Moreover, our exploratory analyses have unveiled nuanced patterns in tract density, motivating further investigation into specific tracts and their intricate connections with cognitive function.

These findings underscore the intricate interplay between white matter connectivity and cognitive states, reinforcing the imperative for comprehensive assessments of structural networks in the realm of neurodegenerative disorders. While our study represents a substantial advancement, our work opens up new avenues for future exploration, encompassing broader dataset generalization, additional feature incorporation, and refinement of the classification pipeline. The fusion of advanced neuroimaging tools and computational modeling holds immense promise for unraveling the intricate mechanisms that underlie cognitive states, ultimately enhancing our capacity to effectively address the complexities of cognitive disorders.

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